Erebus 2027

Release and validation record
27.0-draft.1 • Webots R2025a • 3 November 2026
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Repository structure

Branch	Purpose
2026_webots2023b	Unchanged 26.2 edition; 2026 rules and Webots R2023b. Repository default.
2027_webots2025a	2027 draft rules with the customized Match Studio supervisor; Webots R2025a only.
2028_webots2025a	Future robot configuration and passive-body design. Preserved development prototype; no 2028 rule compliance claimed.

The repository remains a fork of robocup-junior/erebus. The 2026 branch retains commit dddb5c52a556f09c663e4a142ba5d67f3e48305f. All 17 original branches and 31 original tags were backed up before cleanup. The 16 obsolete branches were removed only after their exact commits were verified in the private backup.

Competition graphics profile: Ambient Occlusion Low; Texture Quality Low; Max Texture Filtering 4; Shadows Disabled; Anti-aliasing Disabled. Apply through Webots preferences on competition machines.

2027 behavior

• Battery consumption is calculated by the supervisor from measured wheel motion, with configurable sensor and passive rates. Participant game-info replies include battery percentage.

• A continuous stationary stop on the same cyan contact tile replaces the battery after five seconds. Moving, leaving or relocating resets the stop. Empty battery ends the match before another replacement can occur.

• Swamps increase battery consumption instead of accelerating the match clock. Smiley victims, 20 mm victim signs, 50 mm cognitive targets, non-scoring decoys, a 60 mm reporting radius and contact-map code k follow the draft.

• Match Studio controls, source refresh, Keep Files, HUD, map review, event CSV, remote controllers and self-contained interactive replay exports are retained.

Draft decisions

The 8 September 2026 draft specifies five seconds for battery replacement; this overrides the ten-second example in the supplied feedback. Battery calibration is not specified by the draft: the supplied motor calibration is retained, and the organizer-configurable swamp multiplier defaults to five. The explicit map formula 1 + 1.2 × correctness is used, giving a maximum factor of 2.2, despite conflicting prose elsewhere in the draft.

Validation

Check	Result and scope
Native R2025a, clean Linux installation	32 programmed checks passed across the core and focused Docker runs, using the official R2025a package and software rendering.
Native R2025a, this Mac	31 checks passed in one minimized instance: all 14 worlds, actual robot devices, rules, scoring, full simulated match, IPC/TCP and replay exports. No engine errors or warnings. Docker remains qualified on Linux.
Fields and physical scene	All 14 bundled worlds, including the internal test world, load; generated two/four-wheel models, camera-free and custom-camera models, actual camera images, LiDAR ranges and motion-sensor readings are checked.
Competition behavior	Contact timing, movement reset, pause behavior, swamp drain, natural battery depletion, full 480-second simulated match, token scoring, map evaluation and exit/final-score rounding.
Controller connections and exports	Local, IPC, TCP and real Docker controller; live 20-byte game-info replies; replay companion files, W3D scenes and ZIP integrity.
Portable checks	17 regression scripts plus the interface contract. Mac, Linux and Windows checks passed on Python 3.10 and 3.12; local macOS Python 3.10 checks also passed.
Rendered interface and replay	Full-width and 390 × 740 supervisor layout; low-battery state and setup controls; actual R2025a replay rendering, pause and backward/forward seeking without browser errors.

Corrections found during qualification

Incompatible retained robot sources are rejected before simulator insertion, avoiding pending field changes on removed models. Official ring geometry is bundled locally for clean R2025a installs. Generated camera interfaces omit unused fields. Repeated or cross-category token reports cannot award neighboring signs; final map bonuses apply once. Replay label inventories exclude stale labels from earlier worlds.

Limits

This implements a draft, not a finalized 2027 rulebook. Administrative judging procedures remain human responsibilities. The full Mac rerun confirms the retained-model correction without a crash. Its R2025a installation includes the earlier startup-port repair; Linux used the unmodified package. Native Windows Webots and identical physics across hardware are not certified. Programmed fixtures do not demonstrate an autonomous entrant. The 600-second wall cutoff uses a timer-boundary test. UI layout tests use a connection fixture; native tests separately exercise the real supervisor and controllers. Headless CI checks camera output before disabling unused imagers in timing fixtures; the separate sensor cases keep imaging enabled. Physics and competition time limits remain active. These are functional checks, not performance benchmarks.

Sources and recovery

Official 2027 draft • Official 2027 implementation • Battery reference

The maintainer holds a private GitHub backup and a verified local recovery bundle. Preserved designer source